Date:

```
·START
//Procedure to input readings
1) PROCEDURE Input Readings
   (BYREF Temp: ARRAY[1:50, 1:100] OF REAL,
   Sensors : INTEGER, Readings : INTEGER)
2) DECLARE i, j : INTEGER
3) FOR i ← 1 TO Sensors
4) OUTPUT "Enter readings for sensor", i
5)   FOR j ← 1 TO Readings
6)     INPUT Temp[i,j]
7)   NEXT j
8) NEXT i
9) ENDPROCEDURE

// Function to Analyse one sensor individually
10) FUNCTION AnalyseSensor
   (Temp : ARRAY[1:50, 1:100] OF REAL, S : INTEGER,
   Readings : INTEGER, BYREF Total : REAL,
   BYREF Highest : REAL, BYREF Lowest : REAL,
   BYREF Critical : INTEGER, BYREF Warning : INTEGER,
   BYREF Longest : INTEGER) RETURNS REAL
11) DECLARE j, Streak : INTEGER
12) Total ← 0
13) Highest ← Temp[S, 1]
14) Lowest ← Temp[S, 1]
15) Critical ← 0
16) Warning ← 0
17) Streak ← 0
18) Longest ← 0
```

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19) FOR j ← 1 TO Readings
20)   Total ← Total + Temp[S, j]
21)   IF Temp[S, j] > Highest THEN
22)     Highest ← Temp[S, j]
23)   ENDIF
24)   IF Temp[S, j] < Lowest THEN
25)     Lowest ← Temp[S, j]
26)   ENDIF
27)   IF Temp[S, j] >= 40 THEN
28)     Critical ← Critical + 1
29)   ELSE
      ~~ENDIF~~
30)     IF Temp[S, j] >= 35 THEN
31)       Warning ← Warning + 1
32)     ENDIF
33)   ELSE
      ~~ENDIF~~
34)     Streak ← 0
35)   ENDIF
36) NEXT j
37) RETURN Total / Readings
38) ENDFUNCTION
```

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// Finding average sensor
39) FUNCTION HighestAverageSensor (Averages:
    ARRAY[1:50] OF REAL, Sensors : INTEGER)
    RETURNS INTEGER
40) DECLARE i, Best : INTEGER
41) Best ← 1
42) FOR i ← 2 TO Sensors
43)     IF Averages[i] > Averages[Best] THEN
44)         Best ← i
45)     ENDIF
46) NEXT i
47) RETURN Best
48) ENDFUNCTION

// Main Program
49) DECLARE Temp : ARRAY[1:50, 1:100] OF REAL
50) DECLARE Averages : ARRAY[1:50] OF REAL
51) DECLARE Sensors, Readings, i, Critical, Warning,
    Longest, BestSensor : INTEGER
52) DECLARE Total, Highest, Lowest : REAL
53) OUTPUT "Enter number of sensors: "
54) INPUT Sensors
55) OUTPUT "Enter number of hourly readings: "
56) INPUT Readings
```

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57) CALL InputReadings(Temp, Sensors, Readings)
58) FOR i ← 1 TO Sensors
59)     Averages[i] ← AnalyseSensor(Temp, i, Readings,
                Total, Highest, Lowest, Critical, Warning,
                Longest)
60)     OUTPUT "Sensor", i, "Report:"
61)     OUTPUT "Total temperature:", Total
62)     OUTPUT "Average temperature:", ROUND(Averages[i], 2)
63)     OUTPUT "Highest temperature:", Highest
64)     OUTPUT "Lowest temperature:", Lowest
65)     OUTPUT "Critical Readings:", Critical
66)     OUTPUT "Warning Readings:", Warning
67)     OUTPUT "Longest streak (hours):", Longest
68) NEXT i
69) BestSensor ← HighestAverageSensor(Averages, Sensors)
70) OUTPUT "Final Report:"
71) OUTPUT "Sensor with highest average: Sensor ", BestSensor
72) OUTPUT "Its average temperature: ", ROUND(Averages[BestSen-
        or], 2)
73) FINISH
```

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